Assignment 1: Data Analytics (ECMP 5005B)

Fall-2024

Due date: October 6, 2024

Q1: (80 points)

Use the `tq_get()` function in the package **tidyquant** to get the stock prices (from Jan 01, 2018 to December 31, 2022) of the two crypto currencies: Bitcoin ("BTC-USD") and Ethereum ("ETH-USD"). Name your data as **df** and use it to solve the following questions:

1.1) (10 points)

Use the `mutate()` function from the **dplyr** package along with the `ymd()` function from the **lubridate** package to convert the `date` varaible to the format "year-month-day".

1.2) (5 points)

Use the `mutate()` function from the **dplyr** package to create a new variable: $r_t = \log x_t - \log x_{t-1}$, where r_t and x_t are the log returns and closing price series, respectively, at time t. Name the new variables as *ret*. **Hint:** To calculate $r_t = \log x_t - \log x_{t-1}$ for each currency, use the `case_when(symbol == )` function to differentiate between currencies. After computing the returns, apply `drop_na()` function to remove any `NA` values, and then use the `bind_rows()` function to combine the rows from both currencies.

1.3) (10 points)

Use the `group_by()` and `summarise()` functions from the **dplyr** package to calculate the mean and quartiles (1st, 2nd, and 3rd) of the log returns for each currency. Provide a comparison of the mean values and the 1st and 3rd quartiles between the two currencies.

1.4) (10 points)

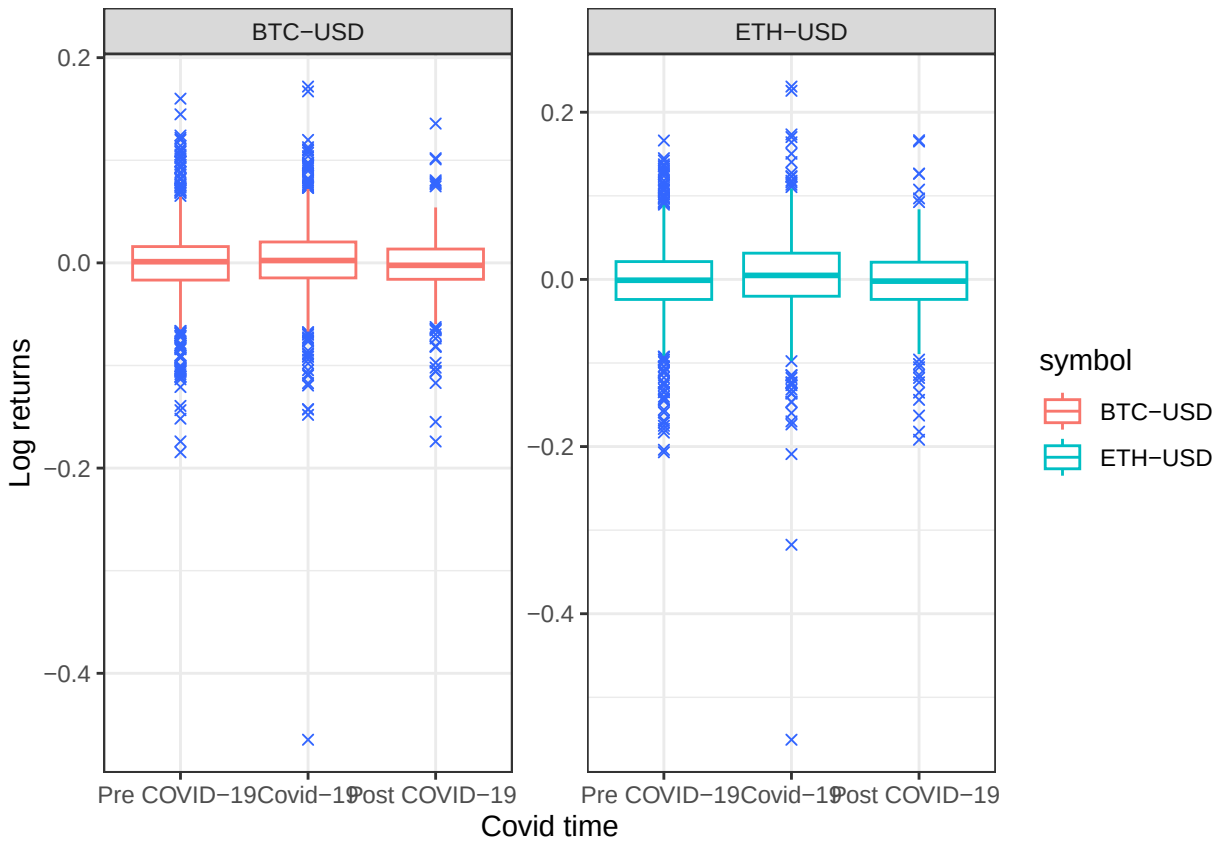
Use the `mutate()` function from the **dplyr** package along with the `case_when()` from the same package and add a new categorical variable called `covid`. This variable should have 3 categories `"Pre COVID-19"` for dates before December 1, 2019, `"Covid-19"` for dates between December 1, 2019, and February 10, 2022, and `"Post COVID-19"` for dates after February 10, 2022.

1.5) (10 points)

Use the **skimr** package to determine the minimum and maximum closing stock prices for both currencies during the COVID-19 period. Compare the minimum and maximum closing prices of Bitcoin during the COVID-19 period with those from the pre-COVID-19 and post-COVID-19 periods.

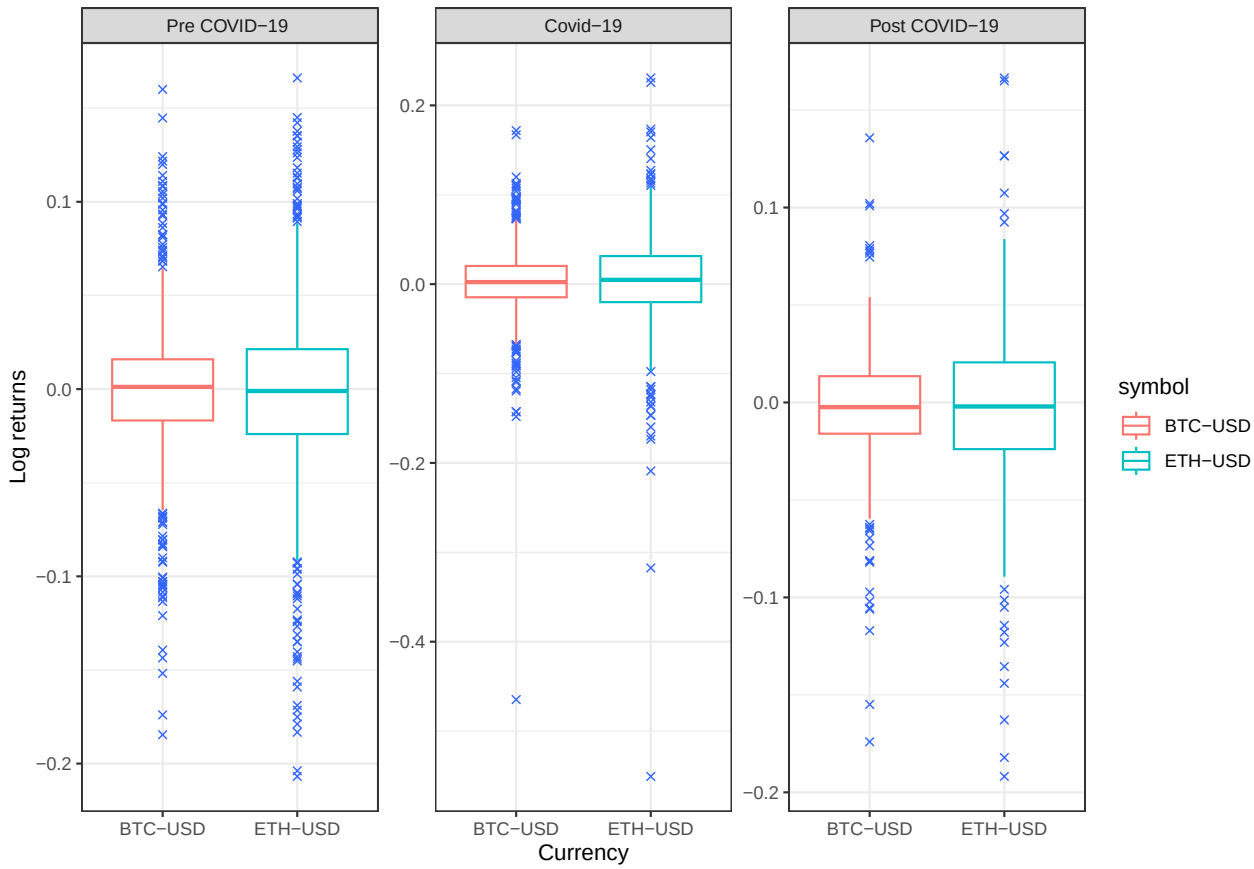
1.6) (10 points)

Use the `facet_wrap()` function from the **ggplot2** package to recreate the plots shown in Figure 1. Write tidy code to generate these plots and provide a commentary on your observations regarding the log returns.



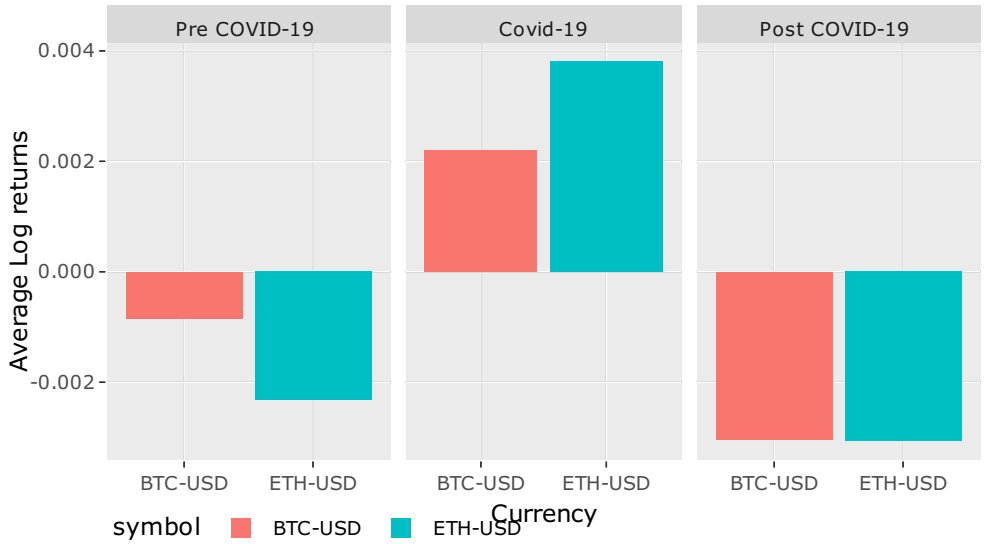
1.7) (10 points)

Write a tidy code to reproduce Figure 2 below.



1.8) (10 points)

Write a tidy code to reproduce the bar chart in Figure 3 below. Note that you need to create an interactive plot using the function `ggplotly()` from the package **plotly**. Comment what do you notice about the log returns for both currencies?



1.9) (5 points)

Use the function `dygraph()` in the package **dygraphs** to produce a dynamic graphic of the returns of the two series. Add a shading effect to your graph background using a `color = "#d3d3d3"` for the covid-19 period.

Q2: (20 points)

Use the `set.seed(123)` and the function `tibble()` to generate a random date of dimension 2000×6 with the following variables and do some summary statistics:

- Gender: (nominal with two categories "male" and "female") 40% are males and 60% are females. Use the `sample()` function.
- Age: (numeric) use gamma-distribution with shape 50. Use the `rgamma()` function.
- Height: (numeric) use normal distribution with mean 160 and standard deviation 8. Use the `rnorm`()‘ function.
- Meal: (integer denotes the number of meals per day) random integers from 0 to 6. Use the `sample()` function.
- Gym: (integer denotes the number of hours spent at the gym per day) use uniform distribution that ranges in value from 0 to 10 and round your generated data to integer numbers. Use the `runif()` function.
- Status: (nominal denotes the marital status) single, married, and others. Use the `sample()` function.
- Weight: (numeric) use the following equation:

```
Weight = 0.06  age + 0.7 height + 0.2 meal - 0.1 male - 0.16 gym - 0.12 single - 0.05
married + random error from t-distribution with 5 degrees of freedom
```

2.1) (10 points)

Write the code that generates the data as described above.

2.2) (5 points)

Find the age of the youngest single male who spends 3 hours at gym per a day.

2.3) (5 points)

Suppose you need to provide a promotion for married customers aged between 40 and 60 who never go to the gym. Write a code which creates a new binary variable (name it as `promotion`) with two categories `yes, no`. What is the weight average of the those who will be eligible to get the promotion?